

```
#include <iostream> using namespace std;
```

```
int binarySearch(int A[], int v, int p, int r) {  
    if (p > r) {  
        return -1;  
    }
```

```
    int j = p + (r - p) / 2;
```

```
    if (v == A[j]) {  
        return j;  
    } else if (v < A[j]) {  
        return binarySearch(A, v, p, j - 1);  
    } else {  
        return binarySearch(A, v, j + 1, r);  
    } }
```

```
int main() {  
    int A[] = {2, 5, 8, 12, 16, 23, 38, 56, 72, 91};  
    int n = sizeof(A) / sizeof(A[0]);  
    int v;
```

```
    cout << "Array: ";  
    for (int i = 0; i < n; i++) {  
        cout << A[i] << " ";  
    }  
    cout << endl;
```

```
    v = 23;  
    int result = binarySearch(A, v, 0, n - 1);  
    if (result != -1) {  
        cout << "Value " << v << " found at index " << result << endl;  
    } else {  
        cout << "Value " << v << " not found" << endl;  
    }
```

```
    v = 50;  
    result = binarySearch(A, v, 0, n - 1);  
    if (result != -1) {  
        cout << "Value " << v << " found at index " << result << endl;  
    } else {  
        cout << "Value " << v << " not found" << endl;  
    }
```

```
    return 0; }
```